

Netflix Movies Recommendation System

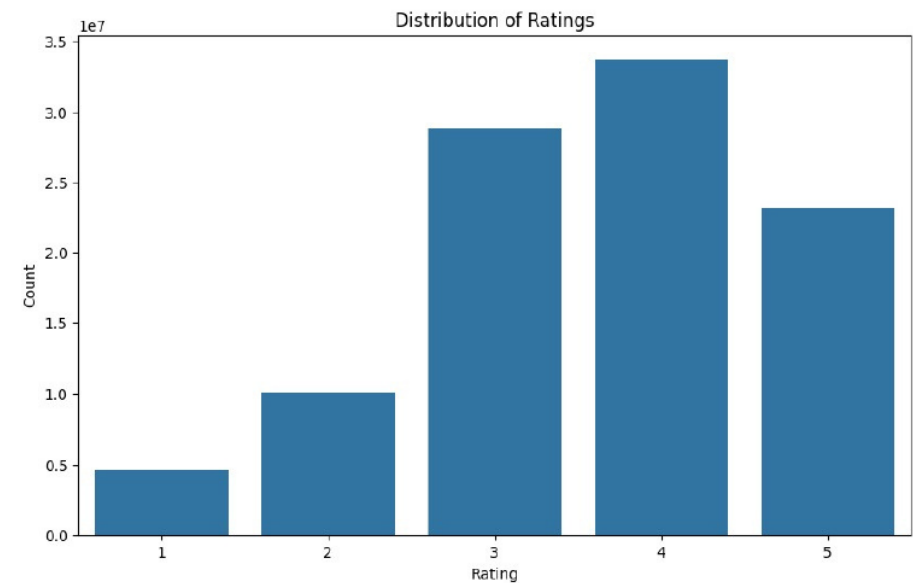
From raw ratings to personalized predictions
Bartosz Maj

Dataset

Netflix Prize dataset containing millions of user–movie ratings.

Key properties:

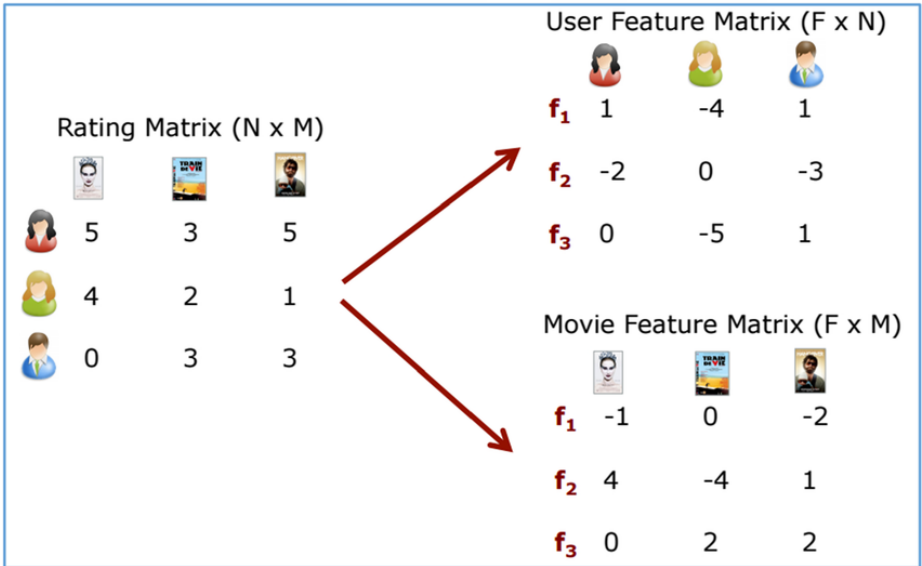
- Extremely sparse matrix
- Ratings from 1 to 5
- Strong long-tail effects



Most ratings concentrate around 3–4.

Model Concept

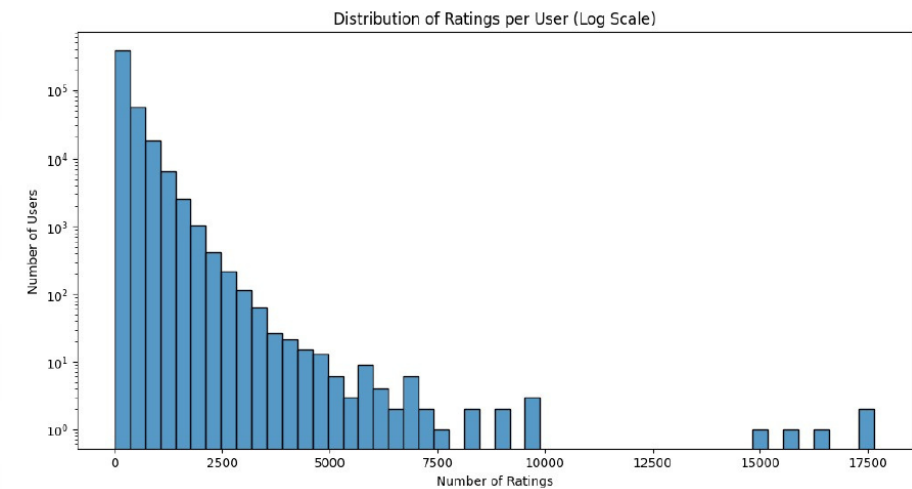
Matrix Factorization decomposes the user–movie matrix into latent preference spaces.



Latent factors capture hidden preferences.

User Activity

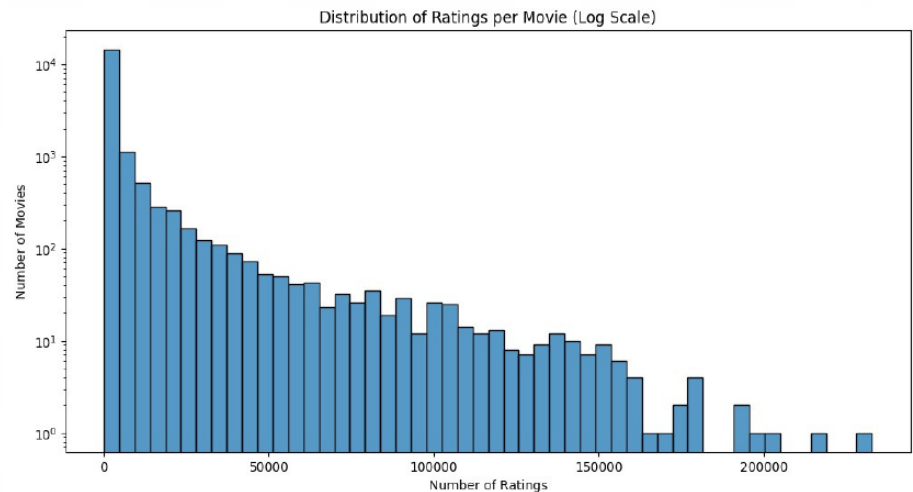
User engagement is highly imbalanced.



Few users generate most ratings (log scale).

Movie Popularity

Movie popularity follows a strong long-tail distribution.

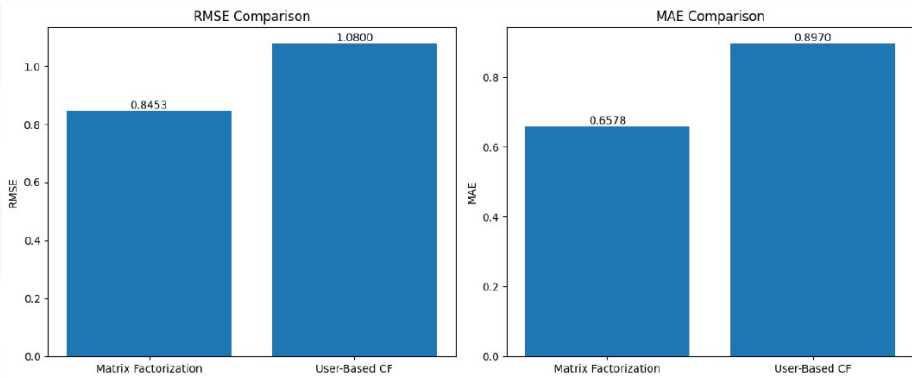


Most movies receive very few ratings.

Baseline vs Proposed Model

User-Biased Collaborative Filtering is used as a baseline. It corrects for individual rating habits by predicting values close to each user's average, but does not model deeper taste structure.

Matrix Factorization learns shared latent preference patterns across users and movies, enabling similarity-based personalization.



Lower RMSE and MAE indicate improved prediction accuracy.

Key Insights

- Baseline corrects rating bias but lacks personalization
- Matrix Factorization captures global preference structure
- Latent factors significantly improve accuracy

What Does the Model Learn?

The model learns hidden dimensions of user taste and movie characteristics, allowing accurate predictions even for unseen user–movie interactions.

Key Takeaway

Effective recommendation is not about predicting average ratings, but about uncovering latent preference structures shared across users and items.

Recommendation Pipeline

Data Preparation → Rating Matrix → Baseline vs Latent Model → Evaluation (RMSE / MAE)